

Enabling AI-Driven VR Theatre: A Low-Latency VR-AI Director Framework with Consistent Personae via Machinery Narrative Audit

Jiayi Wu*
HKUST (GZ)

Sheng Mai†
University College London

Ka Yan Wong
Independent Researcher

Yuyang Wang‡
HKUST (GZ)

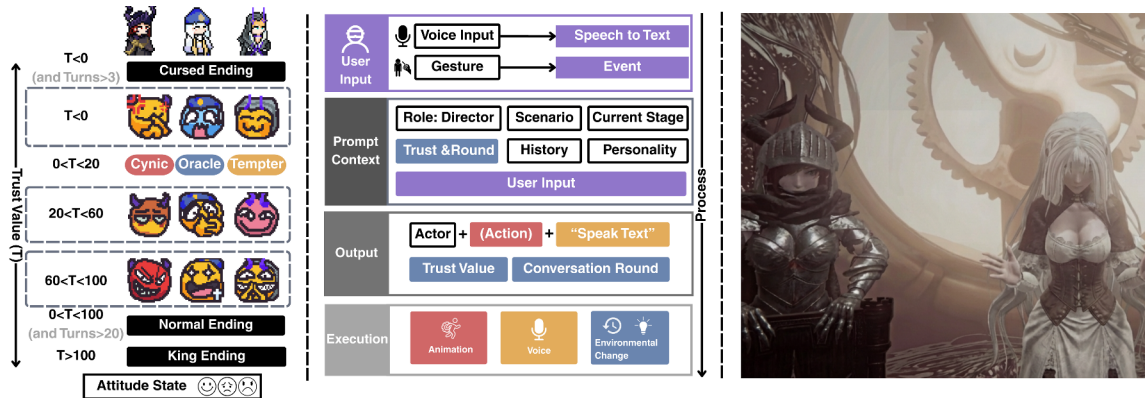


Figure 1: System Overview: (a) Structure of multi-stage narrative progression, (b) VR-AI Director Agents pipeline, (c) VR theatre

ABSTRACT

Location-based immersive theatre faces scalability challenges. VR offers a solution, yet current systems fail to simultaneously achieve low-latency multimodal responsiveness, narrative coherence under user agency, and character consistency across extended improvisation with Embodied Conversational Agents (ECAs). We present a VR-AI Director framework deployed in EveryMacbeth, a VR re-staging of Shakespeare’s Macbeth. The framework integrates (1) parallel scheduling reducing response latency by 81.6%, (2) a Trust Value-driven Procedural Narrative Mechanism, and (3) a Machinery Narrative Audit protocol that reduced persona drift from 8/15 to 2/15 significant shifts through iterative prompt refinement—enabling scalable, coherent VR immersive theatre.

Index Terms: Virtual Reality, Large Language Models, Interactive Narrative, Immersive Theatre.

1 INTRODUCTION

Location-based immersive theatre grows in popularity, exemplified by award-winning Punchdrunk’s Sleep No More (2011) [4] (over 200M ticketed viewers), but the genre faces scalability issues due to venue limitations and production costs.

Virtual Reality (VR)-based theatre becomes a scalable solution, but the current VR-based theatre is restricted to three concurrent requirements: (1) Low-latency multimodal responsiveness (voice, gesture, visual feedback), (2) Coherent and structured narrative under user agency, and (3) Character Consistency (“The Actor’s Mask”) throughout extended improvisational dialogue [5].

*j.wu0420235@arts.ac.uk, Jiayi Wu is also affiliated with the University of the Arts London.

†Jiayi Wu and Sheng Mai contributed equally to this work.

‡Corresponding author.

§e-mail: yuyangwang@hkust-gz.edu.cn

To form a immersive experience, the theatre requires user-determined interaction [2]. No existing VR-AI system or research simultaneously addresses the essentials for VR-based immersive theatre within acceptable latency constraints, with multiple responsive Embodied Conversational Agents (ECAs) [1, 7]. We present a protocol to address the gaps:

- 1. Low-Latency VR-AI Director Framework:** model-agnostic parallel scheduling to coordinate multiple ECAs in multimodal interaction.
- 2. Procedural Narrative Mechanism:** preserve dramatic structure while enabling user autonomy.
- 3. Machinery Narrative Audit:** a pre-deployment visualized protocol to assess ECA persona distinctiveness and stability to evaluate AI actor performance.

2 METHODOLOGY

2.1 Framework Design

EveryMacbeth is a AI-driven VR re-staging of the Witches’ Prophecy scene from Shakespeare’s Macbeth. Players embody Macbeth and interact with three Witch ECAs through speech and gesture (Fig. 1b). This VR-AI Director framework comprises an orchestration layer and an execution layer.

The pipeline triggers from player voice or gesture inputs, with speech transcribed through speech to text model. Subsequently, the Director Orchestration Unit (DOU) employs an LLM as an ensemble manager to synchronise ECA dialogue, procedural animations, and environmental states. To ensure low latency, the system utilises end-to-end streaming: the DOU real-time parses LLM output, allowing action-keywords (e.g., Oracle[raise hand]: “Greetings...”) to dispatch animations and environmental update instantly while each ECA’s dialogue is concurrently synthesised via TTS. The execution layer is a bridge module to deliver synchronised outputs to Unity Engine, which enables parallel multi-ECA responses and a model-agnostic pipeline that eliminates sequential bottlenecks.

2.2 Narrative Design

The narrative follows a stage-based progression governed by ‘Trust Value’, a metric to quantify player-ECA rapport (Fig. 1a). Ini-

tialised at zero, Trust fluctuates (−5 to +5) based on player conduct in each conversational turn: positive input increases value, while hostility decreases it. Reaching stage-specific thresholds (e.g., Trust > 20 for Stage 2) triggers a system transition, updating personae, scene, and environment prompts to alter ECA attitudes and objectives. This ultimately dictates the player’s final ending.

3 EVALUATION AND RESULT

The contribution and effectiveness of the Machinery Narrative Audit protocol are validated in two measurements: (1) the system’s latency, and (2) ECAs persona stability and distinctiveness.

3.1 Latency Performance

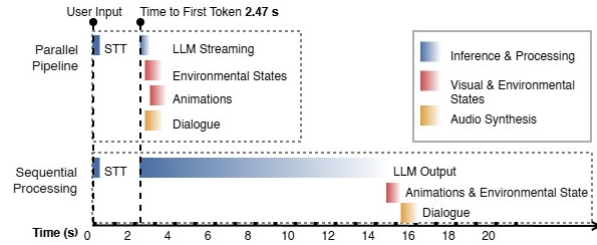


Figure 2: parallel scheduling.

To validate our parallel scheduling pipeline, we benchmarked three LLMs and selected Gemini-3-flash-preview for its lowest first-token latency ($M = 2.47$ s, $SD = 0.04$, $n = 10$). Using sequential processing, end-to-end latency totalled avg. 16.0 s (Fig. 2). Our parallel pipeline reduced time by 80.5% (First user perceived $M = 3.12$ s, $SD = 0.08$) with multimodal interaction.

3.2 Machinery Narrative Audit

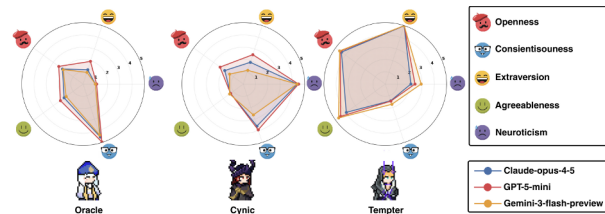


Figure 3: BFI profiles for three witch personas.

One key to dramatic integrity is for each character to maintain a distinct persona and exhibit consistency throughout extended improvisation. We audited the consistency and distinctiveness of the AI-driven characters to assess the stability of their personae by using the 44-item Big Five Inventory (BFI) [3]. This follows recent research utilising the BFI to benchmark LLM personalities and refine prompt strategy [6].

We administered the BFI to each witch persona across three LLMs (Claude-opus-4-5, GPT-5-mini, Gemini-3-flash-preview), and repeated each question ten times. All models produced distinct personality profiles for each witch (Fig. 3), confirming that our persona prompts successfully created differentiable characters.

To test whether these personalities remain consistent in long conversations, we conducted a 20-turn conversation stress test, then compared their BFI from both before and after the session. Paired t-tests ($n = 10$, $p < .05$) signalled significant drift in 8 of 15 personality pairs. We revised the prompts and re-ran the protocol. The number of significant shifts decreased to 2 of 15 (as shown in (Fig.

4). This iterative process illustrates how the Audit functions as a diagnostic tool of persona stability and direct early adjustments.

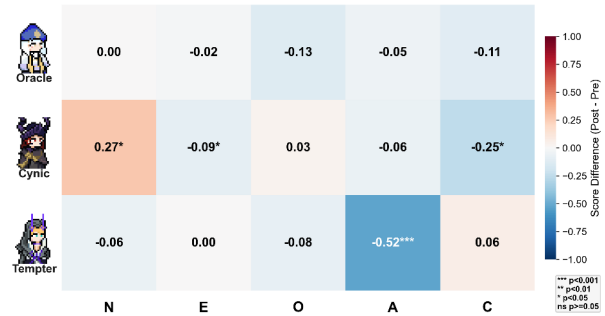


Figure 4: Pre/post BFI difference after prompt refinement.

4 CONCLUSION AND FUTURE WORK

Our protocol provides a technical foundation that addresses key challenges in AI-VR theatre: parallel scheduling facilitates sub-3-second responses, the Trust Value mechanism sustains narrative coherence, and the Narrative Audit visualise and evaluate ECA personality while preserving their consistency.

A limitation is the absence of an audience study with real theatre-goers. We are currently processing issues in comprehensive ethics review, informed consent, and data privacy procedures, as getting approval to conduct a novel AI-VR experience with audience is one of the more strict and long procedures. The Audit protocol establishes a re-deployable pre-deployment standard for AI-driven immersive theatre research. Consequently, future immersive empirical studies and theatre can validate and visualise actor persona consistency before deployment them to audience. Future work will expand this framework to multi-user environments to explore the interplay between social presence and dramatic engagement.

ACKNOWLEDGMENTS

This work is partially funded by the Education Bureau of Guangzhou Municipality, Grant No.2024312045.

REFERENCES

- [1] J. Cassell. Embodied conversational agents: representation and intelligence in user interfaces. *AI Magazine*, 22(4):67–83, 2001. 1
- [2] P. Harrigan and N. Wardrip-Fruin, eds. *Second Person: Role-Playing and Story in Games and Playable Media*. MIT Press, 2007. doi: 10.7551/mitpress/7221.001.0001 1
- [3] O. P. John and S. Srivastava. *The Big Five Trait Taxonomy: History, Measurement, and Theoretical Perspectives*, pp. 102–138. Guilford Press, 2nd ed., 1999. 2
- [4] J. Machon. *Immersive Theatres: Intimacy and Immediacy in Contemporary Performance*. Palgrave Macmillan, 2013. 1
- [5] J. S. Park, J. C. O’Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*, UIST ’23, pp. 1–22. ACM, 2023. doi: 10.1145/3586183.3606763 1
- [6] G. Serapio-Garcia, M. Safdari, C. Crepy, L. Sun, S. Fitz, P. Romero, M. Abdulhai, A. Faust, and M. Mataric. A psychometric framework for evaluating and shaping personality traits in large language models. *Nature Machine Intelligence*, 7:1954–1968, 2025. doi: 10.1038/s42256-025-01115-6 2
- [7] M. Slater. Place illusion and plausibility can lead to realistic behaviour in immersive virtual environments. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 364(1535):3549–3557, 2009. 1